



Pilot Project Proposal

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Lessons Learned: [NMT Lessons Learned Presentation Recording-20260412_190629-Meeting Recording.mp4](#)

Project Objective:

This pilot aims to validate a custom-trained Neural Machine Translation (NMT) engine designed specifically for **Bird & Bird** to streamline the translation of **Patent Applications and IP documents** from English into Spanish.

The core goal is to transition from traditional human translation to a PEMT (Post-editing Machine Translation) workflow that ensures legal compliance while significantly reducing overhead.

Summary of Outcomes:

Based on pilot findings, in which the majority of segments required minimal post-editing and no critical errors were observed, the MT system is expected to significantly reduce post-editing effort, supporting the anticipated increase in throughput.

Recommendations for Further Training:

Based on the eight rounds of training, increasing the size of the training data has been the only reliable way to boost BLEU score. In further rounds of training, more well-cleaned, domain-appropriate data can be introduced to continue improving the BLEU score.

Given the high MQM scores, we recommend introducing machine quality estimation during the training process to minimize human evaluation and save time and money.

Success Criteria:

Efficiency: Achieve an estimated 30–40% increase in translation throughput (words per hour) for patent claims and legal descriptions when using a PEMT workflow.

Methodology: Productivity will be evaluated by comparing the time required to translate content from scratch versus the time required to post-edit MT output.

Cost: Reduce the total cost per word by approximately 30–40% compared to the firm's current standard human translation rates, driven by reduced post-editing effort and increased translator productivity.

Quality Benchmarks (MQM Framework): The post-edited output is expected to achieve an MQM Score of approximately **98/100**. Human evaluation will be conducted at two key stages of the pilot, following Round 1 and Round 8 (post-optimization evaluation). A weighted MQM scoring approach (**Critical = 25, Major = 5, Minor = 1**) will be applied to support quantitative quality estimation. The following quality thresholds will guide evaluation:

Critical Errors (0 Tolerance)	→	Zero mistranslation of patent claims or IP legal terms.
Major Errors (< 2 per 100 segments)	→	Rare omissions that do not alter the core legal meaning.
Minor Errors (< 5 per 100 segments)	→	Minor Spanish syntax or style issues with no legal risk.

Datasets:

Training Data: The pilot will use a legal parallel corpus consisting of approximately 11,210 aligned sentence pairs in English and Spanish, sourced from EUIPO (European Union Intellectual Property Office) trademark opposition and appeal decisions. As the rounds reiterate, segments will be cut shorter in round 2 and 3; more segments will be cleaned and added from round 4-8. The final training dataset will be 12,908 pairs.

Tuning Data: We will be focusing on the client's domain, such as patent claims and IP-related legal language. From round 5-8: tuning set data is cleaned in each round. In round 5 there were 500 segments moved from training to tuning set.

Testing Data: This dataset consists of 500 aligned EN-ES sentence pairs drawn from the same EUIPO trademark opposition and appeal corpus, and will be used to calculate BLEU scores.

Pilot Timeline:

Estimated training time: 6-7 weeks

Potential bottlenecks:

Technical setup - technical difficulties in network environment configuration and system integration

Data cleaning - long hours of sentence alignment for non-TMX corpora

Long training hours due to the size of datasets

Human review - availability of reviewer

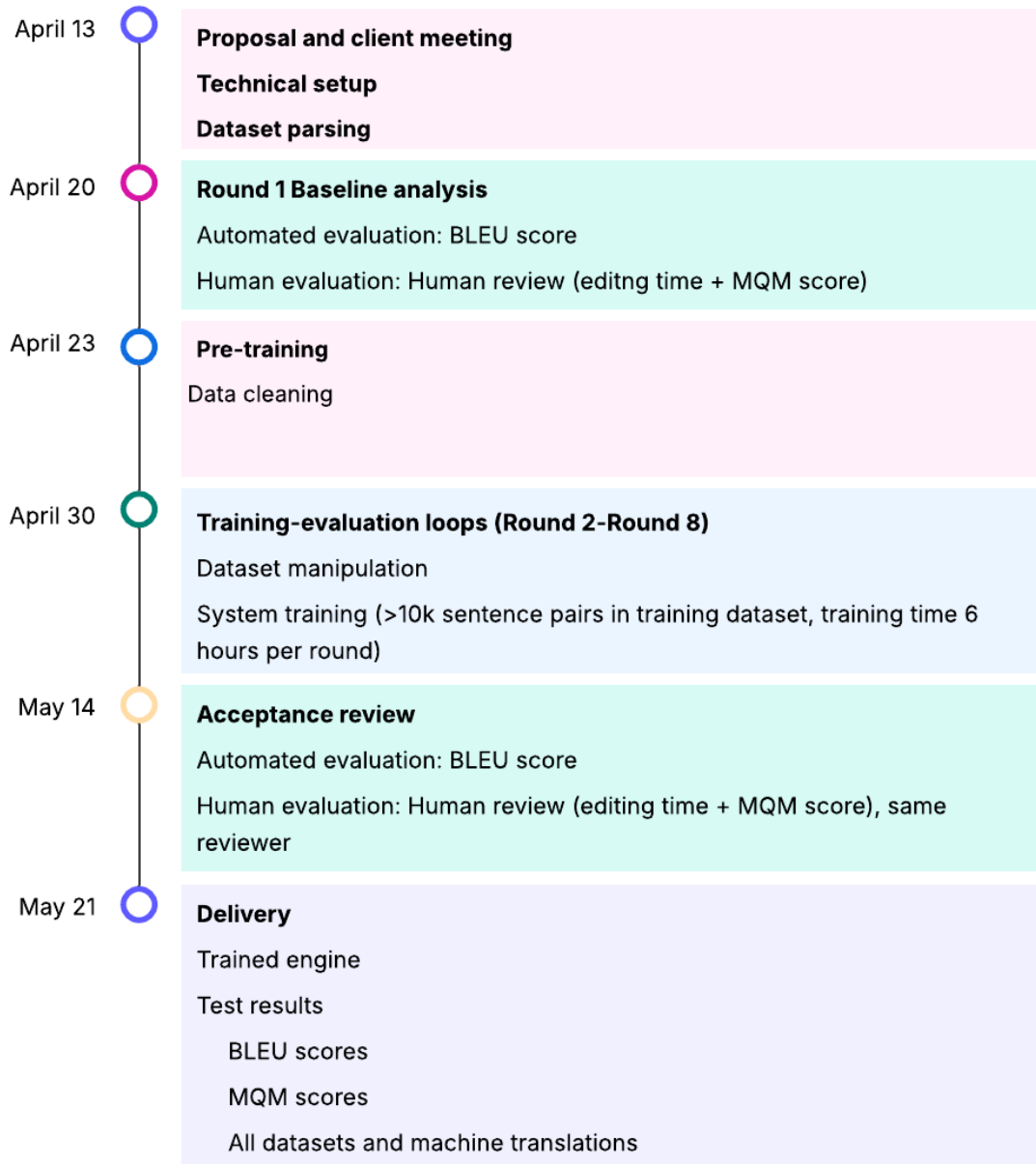
Mitigation in place:

Buffer time included

Experienced engineers on the team

After-hour training - files uploaded at 5pm to allow training in nighttime, saving time for working hours

Translation Engine Training Timeline



Cost:

Role	Task	Date	Hours	# Personnel	Cost per Hour	Total
Engineer	Azure set-up	2-Mar	2	4	\$45	\$360
	Corpus data cleaning	9-Mar	2	2	\$45	\$180
	Round 1*	23-Mar	4	1	\$45	\$180
	Round 2	27-Mar	4	1	\$45	\$180
	Round 3	3-Apr	2	1	\$45	\$90
	Round 4	4-Apr	3.5	1	\$45	\$158
	Round 5	6-Apr	2.5	1	\$45	\$113
	Round 6	7-Apr	2	1	\$45	\$90
	Round 7	8-Apr	2	1	\$45	\$90
	Round 8	10-Apr	2	1	\$45	\$90
Linguist	Human evaluation	28-Mar	4	1	\$40	\$160
PM	Drafting proposal	3-Mar	2	4	18.5% of total project cost	\$310
	NMT training research	9-Mar	2	2		
	General project mngmt		4	1		

Total \$2,000

*Round 1-8 training involved the engineer analyzing the BLEU score from the previous round, designing a training strategy for the next round, manipulating the training and tuning files to create a new training context, and running the files through the training engine.